Issue Brief
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Net Metering in Community-owned Solar Microgrid
A tool for Energy Democracy

The climate crisis not only exacerbates existing social inequalities, but also presents an opportunity to demand systemic change. The root cause of the climate crisis is the dominant fossil fuel-based energy systems most of which emitted by global north countries, this must be destabilized in favor of decarbonized and renewable sources (Strachan et al. 2015). This shift is at the core of the "Energy Democracy" movement, which advocates for a bottom-up approach to energy systems that focus on local autonomy, local participation and ownership. The transition of energy towards renewable has been widely acknowledged and this transition is sustainable, mitigate global greenhouse emission and drive the low carbon economic development. The renewable transition does not only mean moving away from fossil fuel to renewables, but a significant opportunity to address existing structural inequalities and redistribution of the wealth and power associated with it. Furthermore, energy democracy ensures participatory democracy, autonomy and ownership to the communities. The decentralization of energy through community ownership is a viable solution to the concentration of power in the energy sector. By involving the users in the decision-making process, the micro grid and distributed models of grid management can enable local communities to take charge of the production, acquisition, distribution, and consumption of energy (Duda et al., 2017). In addition, we can strive towards creating more equitable and just societies and redistributing power to the people by empowering communities through locally owned renewable energy systems. When structured with a focus on energy democracy and the use of renewable energy, this approach can protect communities from resistance by large utilities and energy interest groups as they work towards building an equitable energy economy.

According to the Climate Justice Alliance'- Energy Democracy represents a shift from the corporate, centralized fossil fuel economy to one that is governed by communities, is designed on the principle of no harm to the environment, supports local economies, and contributes to the health and well-being for all peoples." Thus it connects the transition to renewable energy with a process of democratization, focusing on social benefits and their fair distribution and not on economic advantages for a few.

Community-owned energy is endorsed as an effective way of reducing the political and economic centralization of power in the energy sector and rightly so, as people using the energy would be an active part of the decision-making process. As centralized grid management mostly favors large utilities, it is a major constraint to democratizing the renewable energy sector in the country. In this case the microgrid and distributed models of grid management may serve as an effective tool to build local capacities in the energy sector. If properly structured to focus on energy democracy, decentralization, and distributed renewable energy, it could help shelter local communities and residents build an equitable energy economy from pushback by existing utilities and energy interest groups.

In this case, net metering is perceived as an important energy democracy policy since it makes it possible for diverse, inclusive, community-based generating and ownership models to be implemented. The process of net metering provides system owners with the opportunity to gain extra revenue by selling their excess power to the grid while also making up for shortfalls via the grid. In contrast to electricity generated at large power plants by big companies, this type of system supports the distributed generation of energy where energy comes from decentralized systems.

Net metering is a policy that aligns with the decentralization of energy through community ownership and can help to facilitate the microgrid and distributed models of grid management discussed in the first paragraph. Net metering enables individuals or communities who generate their own electricity, typically using renewable energy sources such as solar panels, to sell excess electricity back to the grid and use the grid as a backup source of power when their own generation falls short. This helps to promote the adoption of renewable energy and can also provide financial benefits to system owners. In addition, net metering supports community-based generating and ownership models by allowing system owners to sell excess power and access the grid as

needed. This is in contrast to traditional electricity generation at large, centralized power plants owned by major companies, and instead promotes decentralized generation of energy from smaller, distributed systems.

Energy democracy has emerged as a means of promoting social justice, particularly in developed nations where there is a growing sense of urgency about climate change caused by extractive economic practices in the global north.

In the United States, distributed energy generation, particularly through solar power, has grown significantly, but opportunities for collective ownership have been limited. While most states have net metering legislation, rules for sharing electricity from non-utility projects are often restricted. Only 16 states currently have policies that allow for electricity sharing, as utilities and corporations frequently work to weaken traditional net metering laws. However, there have been several efforts to advance energy democracy, such as the Stafford Hill Solar Farm in Vermont, which is a fully solar and battery-powered microgrid that provides reliable, green energy to the community while also generating revenue through cost savings and the sale of excess energy. In Boulder, Colorado, efforts to municipalize the city's electrical infrastructure in order to prioritize sustainable energy have faced opposition from corporate interests, but are ongoing. In addition, as many American homes and businesses cannot accommodate solar panels, options for community ownership or shared benefits will be necessary to further increase citizen ownership.

In European countries, hybrid electrical cooperatives, in which owners are both power producers and consumers, have emerged. In Germany, citizens and cooperatives own 50% of the country's renewable energy capacity, and the transition to renewable energy has been described as a democratic movement with citizen ownership at its foundation. In Denmark, community-level cooperatives, often made up of landowners and farmers, own a significant portion of the country's wind turbines and generate a large proportion of Denmark's total energy needs. As a result, 20% of Denmark's energy is generated by wind, and 85% of this is owned by Danish community members. For developing nations, India can serve as a useful case study for net metering policy and implementation, as most states in India offer incentives such as generation-based incentives, exempted banking charges, cross-subsidy charges, electricity duty, and other surcharges.

While social justice is the primary motivator of energy democracy, in developed nations it has emerged in the context of an increasing feeling of urgency regarding global anthropogenic climate change (Burke and Stephens, 2017) triggered by an extractive economy in the global north. In the United States, one of the largest democracies in the world, there has been massive growth in distributed energy generation through the solar, but limited opportunity for collective ownership. Only 16 states now have a policy that permits electricity sharing, as utilities and corporates have frequently fought to weaken traditional net metering legislation. Many initiatives in the path of energy democracy have been attempted including the Stafford Hill Solar Farm in Vermont

Community Owned Dhapsung Solar Micro Grid which is a fully solar and battery-powered microgrid that offers Rutland reliable green energy while making money from cost reductions and the sale of extra energy (Stephens et al., 2018). Despite significant corporate money spent by the existing provider in opposition, Boulder, Colorado's endeavor to municipalize the city's electrical infrastructure to more quickly prioritize sustainable energy is still in progress (Duda et al., 2017). In addition, since almost half of American homes and businesses are unable to host their solar panels, options for community ownership or shared benefits will be necessary to support a further increase in citizen ownership.

In Latin America and the Caribbean, 17 countries have adopted policies to introduce net metering by 2018, with different stages of implementation as pilot, regional, sectorial or national projects (Mejdalani et al., 2018). Energy efficiency is at the core of the Chilean government's energy strategy including the development of distributed generation, smart metering technologies (focusing on Net Metering) and smart grids as a target. The "Los Almendros" suburb, in the area of Huechuraba in Santiago is an example of smart grids where the consumers feel they can now take decisions on their consumption, because they are more informed about energy efficiency (Nigris and Coviello, 2012). In West, South and East Africa shared solar subscription or micro-grid framework has been playing exemplary role in the community with no electricity access and supply. In Mali, the town's school, central administrative building, mosque, health centre and town hall are all now operating on shared solar power and the introduction of shared solar system has brought economic development, and women entreprneurship. Similary, an off-grid solar power system in Remu, Ethiopia, achieved the unprecedented by providing solar power to over 10,000 locals for the cost of just US$2 per person. (Adadevoh, 2017). In many South Asian countries, decentralized renewable energy systems have helped many villages that were deprived of electricity access to produce their own energy and create new opportunities for communities to thrive through job opportunities and entrepreneurship.

For instance, in India, the solar mini-grids in Jharkhand's Gumla district have enabled villagers to access electricity at lower rates than grid-connected power, empowering women-led enterprises and self-help groups to start small-scale rural businesses that run on solar energy. This has not only increased their income but also provided job opportunities and entrepreneurship. Similarly, in the remote settlements of Zanskar in Ladakh, where locals live in extreme climatic conditions and have been struggling to meet their basic needs including electricity, solar micro-grids have made life easier for them. In Bihar and Madhya Pradesh, a community-led solar energy system has achieved 100 percent access to electricity through low-cost, decentralized, micro-grid lighting systems, which has resulted in the creation of youth and women's solar entrepreneurship, youth technicians, and women's self-help microfinance groups in 5 poorest villages. Bangladesh is also playing an exemplary role in this area, with the electrification of 1,199 households on Monpura Island through two solar power mini-grids. This has created new opportunities for jobs and livelihoods for the 684 shops and 41 institutions on the island. Overall, decentralized renewable energy systems have been instrumental in improving the lives of remote communities, promoting entrepreneurship, and reducing poverty in South Asian countries.

In Nepal as well, Microgrids, in particular, have the potential to help the government meet its power needs, especially in rural areas where electricity is not yet accessible. Nepal is determined to provide reliable and clean energy to all its citizens through its Second Nationally Determined Contribution (NDC) targets by 2030. According to the energy component of the

NDC, 5-10% of energy will be generated from renewable sources such as mini and micro hydropower, solar, wind, and bioenergy. The government also plans to create an enabling environment to provide power to small and mid-sized enterprises using distributed renewable energy generation sources.

To support the development of solar energy, the Guidelines for Alternative Electricity Connected to Grid, 2018, have given it special priority. The Mini-grid Special Program Operation Regulations (2076) also require the integration of mini-grids into the national transmission line in coordination with the National Electricity Authority (NEA). Energy producers can sell the excess energy they generate to the national grid through a system known as net metering or net payment. The ministry has classified solar plants into three categories based on their installed capacity. Stations producing 500 watts to 10 kilowatts are called domestic producers, while plants producing 10 to 500 kilowatts are called institutional producers. Producers must apply to the NEA for permission to connect to the national grid and obtain approval from the Department of Electricity Development before installing solar stations. However, NEA had halted the facility of Net metering for a time period and resumed it after receiving pressure from renewable energy promoting institutes. After resuming the service, NEA has set a revised feed-in tariff of Rs. 5.94 per kilowatt-hour for grid-connected solar plants. As of August 2018, 22 MW of solar projects had reached the post-power purchase agreement stage, with an additional 50 MW planned to connect to the national grid.

A case study of Dhapsung village

In 2016, Grid Alternatives in collaboration with Digo Bikas Institute installed a 16kW solar microgrid in Dhapsung village, Helambu Rural Municipality ward no 6, a rural and impoverished village of Sindhupalchowk district in Nepal. Prior to the installation, the village had a 3kW Peltric set for basic lighting, but it was destroyed in the 2015 earthquake. The 16kW solar microgrid is the first of its kind to be owned and operated by a women's group in Nepal, the Dhapsung Women Group. The group was responsible for setting the tariff rates for the electricity generated with input from the users. In addition to providing lighting for 52 households, the community-owned solar microgrid also enabled productive end uses such as electric saws, pumps, and cardamom roasters, improved community engagement through lighting in public spaces, and increased access to information through the use of TVs and phones.

The solar microgrid also had social empowerment benefits for the community. The women of Dhapsung, who had previously been suppressed by patriarchal norms and values, gained decision


Dhapsung Village

making authority in their community as managers and operators of the grid. The income generated from the microgrid also increased the community's access to microloans provided by the Women Group at a 16% interest rate for small businesses or emergencies.

However, the national transmission line was later connected to Dhapsung, and the solar microgrid has been inactive since then. The preference for the national grid as the ultimate solution and alternative systems as temporary measures (Sharma, 2020) has contributed to this shift. Despite frequent and long interruptions in electricity supply from the Nepal Electricity Authority (NEA) (Hashemi, 2021), the presence of the national grid provides a sense of security to the people with minimal participation in operation and maintenance (Sharma, 2020). This has allowed the centralized monopolistic utility to maintain profitability. However, the transformation of the people of Dhapsung from active consumers of the solar microgrid to passive consumers of the national utility has resulted in the loss of socioeconomic benefits from the microgrid. Firstly, the revenue generated from the microgrid that was being reinvested into the local economy through microloans is now lost to the NEA. Secondly, the organizational structure of the microgrid that enabled the Dhapsung Women Group to have decision-making authority is now redundant.

To address this issue, the plan to introduce a net metering facility for the Dhapsung solar microgrid has been endorsed with the agreement of the Women Group and the user group. DBI coordinated with the NEA Melamchi to connect the microgrid to the national grid through the Dhapsung Women Group and Women group had submitted the letter to NEA Melamchi in the process. However, there are several challenges to overcome. There is policy ambiguity regarding the registration of the Women Group as a customer of the NEA, as the microgrid is not currently registered as a customer. Additionally, a circular from the Nepal Electricity Authority dated July 1, 2018 stated that no net payment would be made to consumers if more energy were fed into the grid than consumed. The average power demand at present in the community is low (around 4 kW), so increasing the community's productive end use (PEU) to utilize the full 16 kW capacity of the microgrid is a challenge. Digo Bikas Institute is working with the Dhapsung Women Group to increase the community's PEU and secure the financial benefits of the microgrid, but this requires building the capacity of the community and addressing policy and regulatory barriers. There was no distinct direction provided by the utility authorities and moreover, the role of local government in this process remains ambiguous and needs better understanding. Hence, a distinctive range of queries related to smooth net metering, the role of the municipality in the process and policy vagueness lies ahead.

Dhapsung Women Group can be at the forefront of energy ownership and build community resiliency as a whole by setting a precedent for endorsing community micro grid and Energy Democracy in Nepal. They would be able to decide on the energy

transition process and ultimately ensure the financial benefits stay in the community. The process of localizing the energy system, however, may be challenging as the policy landscape of solar net metering in Nepal is at an emerging stage and conflicting in nature. As presently constituted, there is no provision for clients to get payment from NEA if solar energy production exceeds what a customer uses from the grid. From both a process and policy standpoint, the utility authority lacks clarity. A different spectrum of challenges exists including long lead-time for approval of the net metering agreement, varied year on year subsidy, lack of awareness campaign for the end users, limited awareness of net metering policies and lack of clarity in net metering billing formats (Pokhrel, 2019). This demands a substantial shift in the dialogue, advocacy and policy revision around net metering and energy decentralization issues. In general, policies at the national level are required to permit, support, and even encourage communities to municipalize their energy systems.

The Dhapsung Women Group has the potential to lead the charge for energy ownership and resilience in their community by advocating for the implementation of community microgrids and Energy Democracy in Nepal. By taking control of the energy transition process, they can ensure that the financial benefits of clean energy stay within the community. However, the process of localizing the energy system may be difficult as Nepal's policy on solar net metering is still in its early stages and can be contradictory. There are also other challenges to consider, such as the lengthy approval process for net metering agreements, inconsistent subsidies from year to year, a lack of awareness and understanding of net metering policies among end users, and unclear billing formats. In order to overcome these obstacles, there needs to be a significant shift in the dialogue, advocacy, and policy revision surrounding net metering and energy decentralization. Ultimately, it will be necessary for national policies to allow, support, and even encourage municipalities to own the energy systems.

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Written by: **Arati Poudel and Abhishek Shrestha**

For more information
Battisputali, Kathmandu, Nepal
Phone: +977 1 4610656 |
E-mail: info@digobikas.org
Website: www.digobikas.org